Forum:	Disarmament and International Security Committee (GA1)
Issue:	Assessing the implications of drone warfare for asymmetric conflict in the Red Sea
Author:	Effrosyni Christidou
Position:	Co-Chair

Personal Introduction

Dear delegates,

I am Effrosyni Christidou, and I am a student at the German School of Athens. I have been a part of the MUN community for 4 years, and I consider that the conference organised by Moraitis holds a special place in my heart. I hope that, serving as one of your Co-Chairs in the Disarmament and International Security Committee, I will be able to provide you with equally wonderful experiences as those I have had as a delegate. This will be my 27th conference and my sixteenth time chairing.

The theme "Equity in an Uneducated World" highlights how unequal access to education can deepen existing inequalities. Although education has expanded globally, millions of children and young people remain out of school, while poverty, gender, disability, location and other circumstances can create major barriers to learning. UNESCO reports that 273 million children and youth were out of school in 2024, showing that universal access to education is still far from reality.[1] Therefore, achieving equity requires countries to focus not only on getting children into school, but also on ensuring that those facing the greatest disadvantages receive the support they need.

This study guide should only constitute the basis of your research. I advise that you also look up the issue more thoroughly on your own and try to contrive more solutions. Should you have any questions, please do not hesitate to contact me at effrosynichristidou@gmail.com or the Chair of the committee, Dimitris Flerianos, at dimitris.flerianos@gmail.com . I look forward to meeting you all.

Yours sincerely,

Effrosyni Christidou

[1] UNESCO. (2026). *Global Education Monitoring Report 2026: Access and equity*. https://www.unesco.org/reports/gem-report/en/2026

Introduction

The Red Sea is a major maritime corridor connecting Europe and Asia through the Suez Canal, making the security of the region important to international trade. Since late 2023, attacks by the Houthi movement in Yemen have disrupted commercial shipping in the Red Sea and demonstrated how one can use relatively inexpensive technologies to challenge much stronger military and commercial actors.[2] Among other forces in the region, the Houthis have increasingly relied on drones, missiles and unmanned surface vessels as part of their attacks, giving them the ability to threaten vessels while maintaining an asymmetric advantage over more powerful opponents.[3]

The use of drones is particularly significant because it has changed the way asymmetric conflict can be conducted at sea. Research on Houthi drone warfare shows that the group has developed different strategies for using drones to exploit the vulnerabilities of more powerful opponents and achieve strategic effects at comparatively low cost.[4] Their attacks on commercial shipping have also demonstrated that drone warfare can have consequences beyond the immediate battlefield, contributing to changes in shipping routes, increased costs and wider disruption to global trade.[5] Reuters reported that by July 2024, Houthi attacks had included more than 70 incidents involving commercial vessels, with the group expanding its use of explosive unmanned surface vessels alongside aerial drones and missiles.[6] Other regional actors, including Iran, Hezbollah, Hamas and Iran-aligned groups in Iraq, have also increasingly incorporated drones into their military strategies, highlighting the wider spread of drone warfare across the Middle East.

This study guide therefore examines the implications of drone warfare for asymmetric conflict in the Red Sea, focusing on how Houthi drone operations have affected maritime security, international trade and the ability of conventional military powers to respond to attacks by a non-state actor. The case demonstrates how advances in relatively accessible unmanned technologies are changing the relationship between military power, cost and strategic influence in modern asymmetric conflict.[7]

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[2] UNCTAD. (2024). *Navigating troubled waters: Impact to global trade of disruption of shipping routes in the Red Sea, Black Sea and Panama Canal*. https://unctad.org/system/files/official-document/osginf2024d2_en.pdf

[3] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

[4] Armed Conflict Location & Event Data. (2024). *Six Houthi drone warfare strategies: How innovation is shifting the regional balance of power*. https://acleddata.com/report/six-houthi-drone-warfare-strategies-how-innovation-shifting-regional-balance-power

[5] UNCTAD. (2024). *Red Sea crisis and implications for trade facilitation in Africa*. https://unctad.org/news/red-sea-crisis-and-implications-trade-facilitation-africa

[6] Reuters. (2024). *Houthi explosive drone boat attacks escalate Red Sea danger*. https://www.reuters.com/world/middle-east/houthi-explosive-drone-boat-attacks-escalate-red-sea-danger-2024-07-03/

[7] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

Key Terms and Definitions

Asymmetric Warfare

"Unconventional strategies and tactics adopted by a force when the military capabilities of belligerent powers are not simply unequal but are so significantly different that they cannot make the same sorts of attacks on each other"[8] In the Red Sea, asymmetric warfare is utilised by groups such as the Houthis to challenge more powerful military forces through lower-cost tactics, including drone and missile attacks on commercial shipping.

Drone Warfare

"Drone Warfare is a military strategy that utilises Unmanned Aerial Vehicles (UAVs) to engage targets, fundamentally transforming traditional combat dynamics. This approach allows for remote operations, reducing the risk to human soldiers and enabling precise strikes. As technology advances, drone warfare increasingly influences geopolitical strategies and international security policies."[9]

Unmanned Aerial Vehicle (UAV)

"An aircraft that does not have a pilot but is controlled by a person on the ground, used especially for the deployment of bombs, or for surveillance"[10] In the Red Sea, UAVs are utilised by armed groups such as the Houthis to conduct surveillance and attacks on maritime targets, allowing them to threaten shipping while limiting the risks and costs associated with conventional military operations.

Non-State Actors

"Non-state actors (NSAs) are organisations or entities that operate independently of any recognised government, playing significant roles in political, economic, and social spheres around the world. This category encompasses a wide range of entities, including non-governmental organisations (NGOs), multinational corporations, armed groups, and social movements, among others."[11] Non-state actors, particularly the Houthi movement, play a significant role in the Red Sea by using drones and missiles to challenge conventional military powers and disrupt commercial shipping.

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[8] Encyclopaedia Britannica. (n.d.). *Asymmetric warfare*. Encyclopaedia Britannica. https://www.britannica.com/topic/asymmetric-warfare

[9] Gilli, A. (2022). *Drone warfare: An evolution in military affairs*. NATO Defense College. https://www.ndc.nato.int/drone-warfare-an-evolution-in-military-affairs/

[10] Cambridge Dictionary. (n.d.). *Unmanned aerial vehicle (UAV) definition*. Cambridge University Press. https://dictionary.cambridge.org

[11] EBSCO. (2023). *Non-state actor (NSA).* Research Starters: Politics and Government. https://www.ebsco.com/research-starters/politics-and-government/non-state-actor-

The Red Sea

"A narrow sea connected with the Gulf of Aden and the Indian Ocean, the Red Sea separates the Arabian Peninsula from northeastern Africa. With its connection to the Mediterranean Sea via the Suez Canal, it is one of the most heavily travelled waterways in the world"[12]

The Suez Canal

The Suez Canal is an artificial waterway in Egypt that connects the Mediterranean Sea to the Red Sea, providing a vital shipping route between Europe and Asia. The Suez Canal extends from Port Said on the Mediterranean in the north to the Port of Suez on the Gulf of Suez in the south. [13]

Background Information

Historical Background

Historically, the Red Sea supported coastal communities through fishing, local exchange, port activity and maritime trade, while also connecting wider commercial networks between Africa, Arabia, the Mediterranean and the Indian Ocean.[14] However, this changed with the opening of the geopolitically significant Suez Canal in 1869, which transformed the waterway into a vital corridor linking the Mediterranean Sea with the Indian Ocean. [15]

Throughout the 20th century, the Suez Canal shaped the Red Sea's global relevance because it provided a shorter maritime route between Europe and Asia, making the region strategically important for international trade and energy transportation[16] and it was a recurring flashpoint for conflict, most notably during the 1956 Suez Crisis and the 1967 Six-Day War between Arab states and Israel. Before the canal was opened in 1869, ships travelling between Europe and Asia had to sail around the Cape of Good Hope at the southern tip of Africa, making the journey significantly longer and increasing both travel time and transportation costs. The 1956 Suez Crisis began when Egyptian President Gamal Abdel Nasser nationalised the Suez Canal, leading Britain, France and Israel to launch a military intervention against Egypt. International pressure forced the three countries to

[12] Encyclopaedia Britannica. (n.d.). *Red Sea*. Encyclopaedia Britannica. https://www.britannica.com/place/Red-Sea

[13] State Information Service. (n.d.). *Suez Canal*. Egyptian Economy Sectors. https://sis.gov.eg/en/economy-investment/economy/egyptian-economy-sectors/suez-canal

[14] Blue, L., Thomas, R., & others. (2018). *The archaeological record of Indian Ocean engagements in the Red Sea*. Oxford Research Encyclopedia of Archaeology. https://academic.oup.com/edited-volume/43506/chapter/364132532

[15] Suez Canal Authority. (n.d.). *History of the Suez Canal*. https://www.suezcanal.gov.eg/English/About/Pages/History.aspx

[16] U.S. Energy Information Administration. (2024). *The Suez Canal and SUMED Pipeline are critical chokepoints for global oil and natural gas transit*. Today in Energy. https://www.eia.gov/todayinenergy/detail.php?id=61984

withdraw.[17] In 1967, during the Six-Day War, Israel fought Egypt, Jordan and Syria and captured several territories, including Egypt's Sinai Peninsula. Egypt subsequently closed the Suez Canal, disrupting international shipping until it reopened in 1975.[18]

After the Cold War, international attention increasingly shifted toward the southern Red Sea, particularly the Bab al-Mandeb Strait and the Gulf of Aden, as concerns over terrorism, piracy and weak state institutions grew. Yemen's strategic position along the Bab al-Mandeb made its internal instability particularly important because conflict on land could directly affect maritime security and international shipping.[19]

The 2011 Arab Spring further exposed Yemen's political fragility, as widespread protests contributed to the departure of President Ali Abdullah Saleh and an unsuccessful political transition under his successor, Abd Rabbu Mansour Hadi.[20] In 2014, the Houthi movement seized control of Sana'a, and the conflict escalated in 2015 when a Saudi-led coalition intervened in support of the internationally recognised government.[21] This transformed Yemen's internal conflict into a broader regional struggle involving Saudi Arabia, Iran and other external powers, while also creating conditions in which armed groups could exploit the country's instability.[22] As the conflict continued, the security of the Bab al-Mandeb became increasingly connected to Yemen's civil war, demonstrating how political instability on land can create wider maritime and international security consequences.[23]

The Construction of the Suez Canal

The Suez Canal presents an excellent instance where the difficulties prompted the discovery of means to overcome them. The work was fortunate in having as chief contractors two men of extraordinary energy and resources, Borel and Lavalley, who took over the work in 1865.[24] They recognised that the

[17] U.S. Department of State, Office of the Historian. (n.d.). *The Suez Crisis, 1956*. https://history.state.gov/milestones/1953-1960/suez

[18] U.S. Department of State, Office of the Historian. (n.d.). *The Arab-Israeli War of 1967*. https://history.state.gov/milestones/1961-1968/arab-israeli-war-1967

[19] Council on Foreign Relations. (2018). *Free Flow of Oil, Strait of Hormuz, and Policing International Sea Lanes*. https://www.cfr.org/articles/free-flow-oil-strait-hormuz-and-policing-international-sea-lanes

[20] Council on Foreign Relations. (2023). *Yemen's Tragedy: War, Stalemate, and Suffering*. https://www.cfr.org/backgrounders/yemen-crisis

[21] Council on Foreign Relations. (2026). *Conflict in Yemen and the Red Sea*. https://www.cfr.org/global-conflict-tracker/conflict/war-yemen

[22] World Bank. (2024). *Yemen Faces Mounting Economic Challenges as Conflict Continues, Regional Tensions Escalate*. https://www.worldbank.org/en/news/press-release/2024/10/31/yemen-faces-mounting-economic-challenges-as-conflict-continues-regional-tensions-escalate

[23] World Bank. (2024). *Yemen Faces Mounting Economic Challenges as Conflict Continues, Regional Tensions Escalate*. https://www.worldbank.org/en/news/press-release/2024/10/31/yemen-faces-mounting-economic-challenges-as-conflict-continues-regional-tensions-escalate

[24] WIT Press. (2003). *The Suez Canal*. Transactions on the Built Environment, 65. https://www.witpress.com/Secure/elibrary/papers/MH03/MH03011FU.pdf

contract could only be achieved with the aid of machinery. They accordingly devised those extraordinary dredgers to suit such difficulties. The dredgers varied in size according to the work for which they were required, and the disposal of the dredged materials.

The practical breadth of a canal is the measured breadth at its bottom. When it opened in 1869, the Suez Canal cross-section bottom width measured 72 feet from end to end, while for seventy-seven miles (78% of the total length) it was double the width.

Although construction of the modern Suez Canal began in 1859, its expansion has continued throughout its history. The canal's current cross-section is approximately 15 times larger compared to 1869, reflecting the need to accommodate the growth in size and volume of maritime traffic.[25] Development of the canal region has also included transportation and irrigation infrastructure, including the Ahmed Hamdi Tunnel, El Sheikh Zaid Syphon, El Salam Canal Syphon, and new ports and bridges.[26]

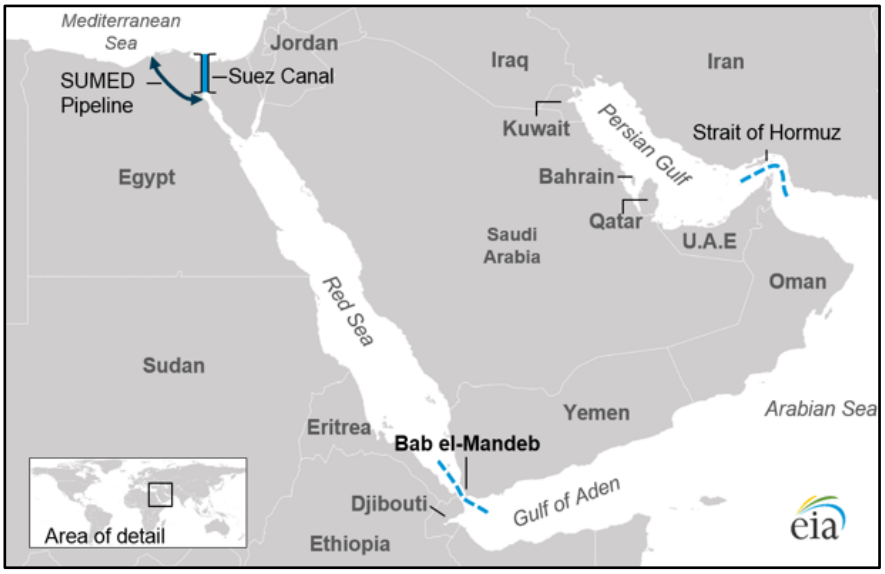


Figure 1: Map depicting the Bab el Mandeb Strait

These continued investments have strengthened the canal's role as a major international trade corridor, allowing it to handle increasingly large vessels and greater volumes of cargo while supporting the economic development of the surrounding region.[27] The canal's importance to global trade was particularly evident in March 2021, when the blockage of the waterway by the Ever Given container ship disrupted shipping and demonstrated how a disruption at a single chokepoint can

[25] Hamza, M. A. (2003). *The construction of the Suez Canal*. Transactions on the Built Environment, 65. https://www.witpress.com/Secure/elibrary/papers/MH03/MH03011FU.pdf

[26] Hamza, M. A. (2003). *The construction of the Suez Canal*. Transactions on the Built Environment, 65. https://www.witpress.com/Secure/elibrary/papers/MH03/MH03011FU.pdf

[27] Suez Canal Authority. (n.d.). *Suez Canal*. https://www.suezcanal.gov.eg/English/About/Pages/default.aspx

have consequences for international supply chains.[28] The six-day blockage halted maritime traffic, delaying hundreds of vessels and disrupting the movement of goods, fuel and other commodities, while also leading to higher shipping costs and supply-chain delays.

The Importance of the Red Sea in Global Trade

The Suez Canal is an important international navigation canal linking the Mediterranean Sea at Port Said and the Red Sea. This way, about 12–15% of global trade passes through the Suez Canal.[29] Due to its location, the Red Sea, as the entrance to the Suez Canal from the south, is one of the busiest shipping routes in the world. The canal provides a significantly shorter route between Europe and Asia, allowing vessels to avoid the much longer journey around the Cape of Good Hope.

The unique geographical position of the Suez Canal makes it of special importance not only to Egypt, but to the entire world, with it providing a much shorter alternative to many maritime trips. This reduction in distance results in shorter travel times, lower fuel consumption and reduced ship operating costs, making the canal an economically efficient route for international shipping.[30] Furthermore, accidents are almost nil compared with other waterways.[31] This importance is getting augmented with the further development of maritime transport and world trade.

The Red Sea has become extremely important economically because it has huge oil reserves and is a source of precious metals. There are large amounts of zinc, copper, silver, gold, and elements such as cadmium, cobalt, and hydrocarbons that have been found in the isolated depths of the Red Sea.[32] Countries such as Saudi Arabia have increasingly developed these resources, particularly gold, copper and zinc, through domestic mining and investment programmes, making mineral extraction an important part of the country's economic diversification strategy. The Suez Canal, which connects the Mediterranean and the Red Sea, handles about 12 per cent of global trade.

Drone Warfare

Drone warfare, or the use of Unmanned Aerial Vehicles (UAVs) in military operations, has developed significantly over the past two decades. Although drones were used for surveillance and reconnaissance in the early 2000s, the development of armed UAVs transformed their role in warfare. The U.S. Predator was equipped with Hellfire missiles in 2001, and armed drone operations soon

[28] UNCTAD. (2021). *Suez Canal blockage: A major disruption to global trade*. https://unctad.org/news/suez-canal-blockage-disruption-global-trade

[29] United Nations Conference on Trade and Development. (2024). *Review of maritime transport 2024*. United Nations. https://unctad.org/publication/review-maritime-transport-2024

[30] Suez Canal Authority. (n.d.). *Why Suez Canal?* https://www.suezcanal.gov.eg/English/About/Pages/WhySuezCanal.aspx

[31] Suez Canal Authority. (n.d.). *Importance & Advantages*. https://www.suezcanal.gov.eg/English/About/WhySuezCanal/Pages/ImportanceAndAdvantages.aspx

[32] Laurila, T. E., et al. (2015). *New insights into the mineralogy of the Atlantis II Deep metalliferous sediments, Red Sea*. Geochemistry, Geophysics, Geosystems. https://doi.org/10.1002/2015GC006010

became an important feature of modern military operations.[33] The proliferation of relatively inexpensive drones has also enabled Non-State Actors to develop capabilities that can challenge stronger conventional forces.[34] For example, in the Red Sea, the Houthi movement has used drones alongside missiles and other unmanned systems to attack commercial and military vessels, while Iran has also used and supplied drone technology in the region, demonstrating how relatively accessible technology can create significant security and economic consequences.[35] The group's attacks have contributed to disruption of commercial shipping, while the sinking of vessels including the Magic Seas and Eternity C in July 2025 demonstrated the potential consequences for maritime security.[36] In response, international actors increased efforts to protect commercial shipping, with the EU's Operation ASPIDES mobilising naval resources to assist affected crews and safeguard freedom of navigation in the Red Sea. The Red Sea therefore provides an important example of how drone warfare can strengthen the capabilities of non-state actors and allow weaker groups to impose costs on more powerful states and the international shipping system.[37]

Recent developments

On 14 September 2019, drone and missile attacks damaged Saudi Arabia's Abqaiq oil facilities, demonstrating the growing importance of drones in asymmetric warfare. Before the current Red Sea crisis, the Houthis had already demonstrated their ability to use UAVs and missiles to threaten Saudi Arabia and its critical infrastructure. In November 2021, Houthi forces launched multiple explosive UAVs towards Saudi targets, including facilities in Jeddah, while Saudi forces responded by targeting Houthi weapons and UAV infrastructure in Yemen.[38] These attacks demonstrated an important feature of asymmetric warfare: relatively inexpensive unmanned systems could be used by a non-state actor to threaten highly valuable infrastructure and force a much more powerful state to invest heavily in defensive systems.[39]

This pattern became significantly more important after October 2023, when the Houthis expanded their use of drones and missiles into the Red Sea and Gulf of Aden. The group stated that its attacks

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[33] Franke, U. E. (2017). *Drone Warfare*. Oxford Bibliographies in International Relations. https://academic.oup.com/reference/62360/reference-article-abstract/554547935

[34] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

[35] International Institute for Strategic Studies. (2024). *The Houthi campaign in the Red Sea: An asymmetric challenge*. https://www.iiss.org/

[36] Reuters. (2025). *Six crew rescued, 15 missing after Houthis sink latest Greek ship in Red Sea*. https://www.reuters.com/

[37] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

[38] Reuters. (2021). *Saudi Arabia says it intercepted Houthi drones and ballistic missiles*. Reuters. https://www.reuters.com/

[39] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

were linked to the war in Gaza and began targeting commercial and military vessels associated, in its stated rationale, with Israel. [40]On 19 November 2023, Houthi forces seized the Galaxy Leader, marking a major escalation from attacks on land-based targets to direct disruption of international maritime shipping.[41] The campaign subsequently involved anti-ship ballistic missiles, cruise missiles, explosive UAVs and unmanned surface vessels, demonstrating the Houthis' growing ability to combine different relatively low-cost technologies in maritime attacks.[42]

Iran has also played a significant role in the development and proliferation of drone warfare in the region. The UN reported that the UAV used in the Houthi attack on Tel Aviv in July 2024 was identified by Israeli authorities as a modified Iranian Samad-3, while the UN Panel of Experts has documented Iranian involvement in the transfer and development of unmanned systems connected to the conflict.

Israel has also become directly involved in the use of unmanned systems in the Red Sea conflict. Following the Houthi UAV attack on Tel Aviv in July 2024, Israel carried out air strikes against Houthi-controlled areas around Hudaydah port, marking an escalation in the confrontation between the two sides and demonstrating how drone attacks in the Red Sea have contributed to a wider regional exchange of military action.

[40] Council on Foreign Relations. (2024). *Conflict in Yemen and the Red Sea*. https://www.cfr.org/global-conflict-tracker/conflict/war-yemen

[41] International Maritime Organization. (2023). *IMO condemns seizure of Galaxy Leader*. https://www.imo.org/

[42] International Institute for Strategic Studies. (2024). *The Houthi campaign in the Red Sea: An asymmetric challenge*. https://www.iiss.org/

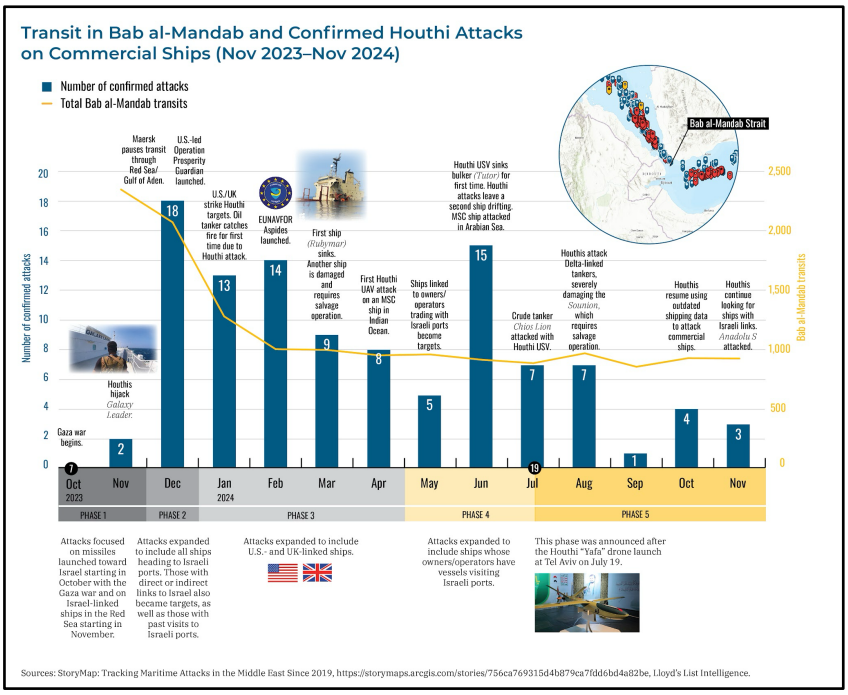


Figure 2 :The graph depicts the transit in Bab al-Mandeb and describes the confirmed Houthi Attacks on commercial ships from November 2023 to November 2024.[43]

The attacks created a security problem that extended beyond Yemen and Israel. Major shipping companies began diverting vessels away from the Red Sea and around the Cape of Good Hope, increasing voyage times and costs and demonstrating how asymmetric attacks by a non-state actor could produce consequences for global commerce.[44] In response, the United States announced Operation Prosperity Guardian in December 2023, a multinational maritime security initiative intended to protect commercial shipping, while the United States and United Kingdom began conducting strikes against Houthi targets in Yemen in January 2024.[45] The Red Sea therefore provides a significant case study of modern asymmetric drone warfare: a non-state actor has used comparatively accessible unmanned technologies to threaten commercial shipping, impose economic costs and compel technologically superior states to deploy expensive military capabilities in response.[46]

Consequences

[43] Raydan, N., & Nadimi, F. (2024). *Houthi shipping attacks: Patterns and expectations for 2025.* The Washington Institute for Near East Policy. https://www.washingtoninstitute.org/policy-analysis/houthi-shipping-attacks-patterns-and-expectations-2025

[44] UNCTAD. (2024). *Navigating troubled waters: Impact to global trade of disruption of shipping routes in the Red Sea, Black Sea and Panama Canal*. https://unctad.org/system/files/official-document/osginf2024d2_en.pdf

[45] U.S. Department of Defense. (2023). *Secretary Austin announces multinational maritime security initiative in Red Sea*. https://www.defense.gov/News/Releases/Release/Article/3618343/secretary-austin-announces-multinational-maritime-security-initiative-in-red-sea/

[46] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

The attacks in the Red Sea through the use of UAVs have created significant human, security and economic consequences, particularly due to their use against commercial and military vessels operating in a major international shipping corridor. Since late 2023, attacks have damaged or threatened vessels from multiple countries, caused deaths and injuries among seafarers, and forced ships to alter their routes to avoid the Bab el-Mandeb Strait.[47] The attacks have also demonstrated the vulnerability of maritime infrastructure and commercial vessels to relatively low-cost unmanned systems, while requiring international naval forces to deploy air-defence and maritime-security capabilities to protect shipping.[48]

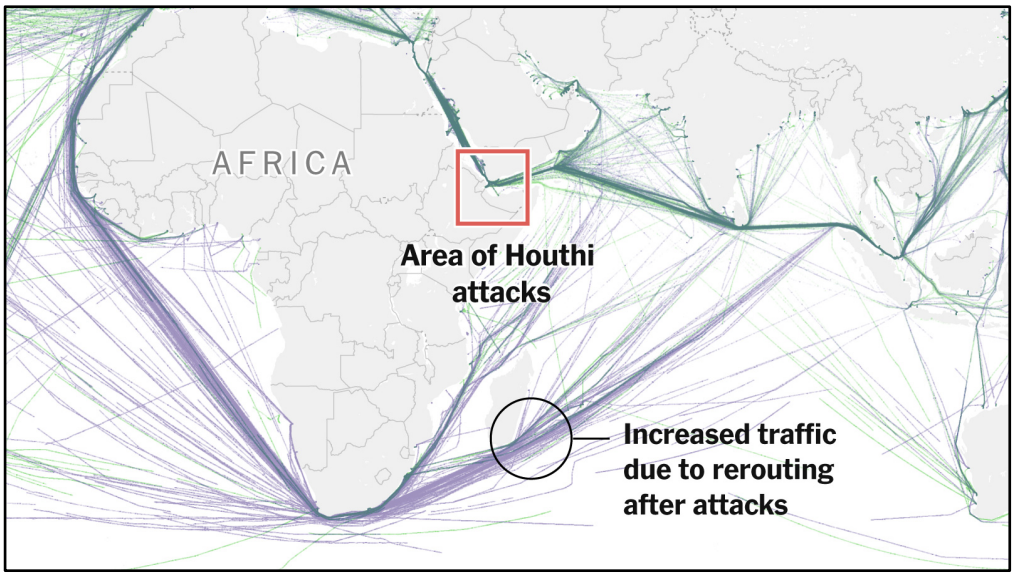


Figure 3: Houthi attacks relative to Africa and maritime transit through the Bab al-Mandeb Strait

Timeline of Events

Date	Description of Event
17 November 1869	Date of completion of the Suez Canal.[49]

[47] International Maritime Organization. (2024). *Red Sea attacks: IMO condemns attacks on merchant ships and seafarers*. https://www.imo.org/

[48] U.S. Department of Defense. (2024). *Operation Prosperity Guardian and maritime security in the Red Sea*. https://www.defense.gov/

[49] Suez Canal Authority. (n.d.). *History of the Suez Canal*. https://www.suezcanal.gov.eg/English/About/Pages/History.aspx

29 October 1888	The Constantinople Convention signed by the United Kingdom, Germany, Austria-Hungary, Spain, France, Italy, the Netherlands, the Russian Empire, and the Ottoman Empire guarantees freedom of navigation through the Suez Canal.[50]
26 March 2015	Saudi Arabia launched Operation Decisive Storm, marking the beginning of regional military intervention against the Houthis in Yemen.[51]
14 September 2019	Drone and missile attacks on Saudi Arabia's Abqaiq oil facilities. Although the Houthis claimed responsibility for the attacks, UN sanctions investigators concluded that it was not Houthi forces that launched the attacks.[52]
7 October 2023	The Israeli-Hamas war began, increasing tensions in the Middle East [53]
19 November 2023	The Houthis seized the cargo ship Galaxy Leader.[54]
December 2023	Operation Prosperity Guardian is announced by the United States and partners to protect Red Sea shipping.[55]

[50] Suez Canal Authority. (n.d.). *The Suez Canal Convention of 1888*. https://www.suezcanal.gov.eg/English/About/Pages/History.aspx

[51] Council on Foreign Relations. (2025). *Conflict in Yemen and the Red Sea*. https://www.cfr.org/global-conflict-tracker/conflict/war-yemen

[52] United Nations Security Council. (2019). *Security Council Briefing on the eighth report of the Secretary-General on the implementation of resolution 2231 (2015).* https://www.unmissions.org/en/dppa/speeches-and-statements/security-council-briefing-eighth-report-secretary-general-implementation-resolution-2231-2015

[53] Council on Foreign Relations. (2024). *Israeli-Palestinian conflict: Gaza war*. https://www.cfr.org/global-conflict-tracker/conflict/israeli-palestinian-conflict

[54] International Maritime Organization. (2024). *IMO condemns "illegal, unjustifiable" attacks on ships in Red Sea*. https://www.imo.org/en/mediacentre/pressbriefings/pages/imo-msc-resolution-red-sea.aspx

[55] U.S. Department of Defense. (2023). *Statement from Secretary of Defense Lloyd J. Austin III on ensuring freedom of navigation in the Red Sea*. https://www.defense.gov/News/Releases/Release/Article/3621110/statement-from-secretary-of-defense-lloyd-j-austin-iii-on-ensuring-freedom-of-n/

Major Actors Involved

Yemen

Yemen is a central actor in the Red Sea conflict. The Houthi movement, which controls significant territory in northern Yemen, including Sana'a, has used the country's territory as a base for launching drones and missiles against targets in the region and against commercial shipping in the Red Sea.[56] The Houthis' development and use of these weapons has allowed them to challenge significantly stronger military powers while imposing wider security and economic costs.[57] Since late 2023, their attacks on vessels in the Red Sea have shown how Yemen's internal conflict has developed into a broader regional security issue, with direct implications for maritime security and international trade.[58] Yemen has experienced a prolonged internal armed conflict, beginning in 2014 when the Houthi movement seized control of Sana'a and later expanding into a wider regional confrontation. Since late 2023, the conflict has also directly affected the wider region, with Houthi attacks on vessels in the Red Sea prompting international military responses and increasing the internationalisation of Yemen's war.

Saudi Arabia

Saudi Arabia is a major actor in the Red Sea's drone-warfare environment. It has both used air power against the Houthis and been repeatedly targeted by Houthi drones and missiles.[59] Saudi Arabia has conducted airstrikes against Houthi military infrastructure in Yemen as part of its wider intervention in the Yemeni conflict, while also deploying air-defence systems to intercept incoming drones and missiles. The Houthis have repeatedly targeted Saudi territory and critical infrastructure, including airports and energy facilities, demonstrating how drone warfare has enabled attacks against strategically important sites far beyond the immediate battlefield. Since 2015, the Saudi-led coalition has conducted military operations against the Houthis in Yemen, while the Houthis have increasingly used UAVs and missiles to target Saudi territory and critical infrastructure.[60] The 2019 attacks on Saudi Arabia's Abqaiq oil-processing facility and Khurais oil field demonstrated the strategic potential of drone warfare, as the attacks temporarily disrupted a significant portion of Saudi oil production and highlighted the difficulty of defending critical infrastructure against coordinated

[56] Council on Foreign Relations. (2024). *Conflict in Yemen and the Red Sea*. Council on Foreign Relations. https://www.cfr.org/global-conflict-tracker/conflict/war-yemen

[57] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

[58] UNCTAD. (2024). *Navigating troubled waters: Impact to global trade of disruption of shipping routes in the Red Sea, Black Sea and Panama Canal*. United Nations Conference on Trade and Development. https://unctad.org/system/files/official-document/osginf2024d2_en.pdf

[59] Council on Foreign Relations. (2025). *Conflict in Yemen and the Red Sea*. Council on Foreign Relations. https://www.cfr.org/global-conflict-tracker/conflict/war-yemen

[60] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

attacks involving UAVs and missiles.[61] The attacks had major economic and strategic consequences, temporarily halving Saudi Arabia's oil production and removing around 5% of global oil supply from the market. Although Saudi production was largely restored within weeks, the attacks caused a sharp rise in oil prices and exposed the vulnerability of critical energy infrastructure to relatively low-cost drone and missile attacks.Saudi Arabia's experience therefore demonstrates how drones can allow a non-state actor to threaten high-value infrastructure and impose high economic and security costs on a technologically superior state.[62]

United States of America

The United States became a major actor in the Red Sea crisis as Houthi forces increasingly used drones, missiles and other unmanned systems to threaten commercial and military vessels.[63] In December 2023, the United States established Operation Prosperity Guardian, a multinational maritime initiative intended to protect shipping and freedom of navigation in the Red Sea and Gulf of Aden.[64] Operation Prosperity Guardian was partially effective in strengthening the protection of commercial vessels and providing a sustained international naval presence, but it was unable to fully deter Houthi attacks or restore normal levels of shipping through the Red Sea, with many vessels continuing to use alternative routes. In January 2024, U.S. and British forces also carried out strikes against Houthi military targets, including missile, radar and UAV capabilities.[65] Although the strikes successfully damaged Houthi missile, radar and UAV infrastructure, they did not stop the group's attacks, proving the difficulty of using conventional military power to fully deter a non-state actor employing relatively low-cost asymmetric tactics. The U.S. response has therefore combined defensive maritime operations with strikes intended to reduce the Houthis' ability to conduct further attacks.[66] However, the continued Houthi attacks after these operations suggest that, while U.S. and coalition forces have successfully intercepted numerous drones and missiles and protected some

[61] U.S. Energy Information Administration. (2019). *The Saudi oil attacks: Abqaiq and Khurais*. Today in Energy. https://www.eia.gov/todayinenergy/detail.php?id=41393

[62] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

[63] U.S. Department of Defense. (2024). *At Gulf Cooperation Council, U.S. participates in maritime, missile defense working groups*. U.S. Department of Defense. https://www.defense.gov/News/News-Stories/Article/Article/3782973/at-gulf-cooperation-council-us-participates-in-maritime-missile-defense-working/

[64] U.S. Department of Defense. (2023). *Statement from Secretary of Defense Lloyd J. Austin III on ensuring freedom of navigation in the Red Sea*. U.S. Department of Defense. https://www.defense.gov/News/Releases/Release/Article/3621110/statement-from-secretary-of-defense-lloyd-j-austin-iii-on-ensuring-freedom-of-n/

[65] U.S. Department of Defense. (2024). *U.S., partners' forces strike Houthi military targets in Yemen*. U.S. Department of Defense. https://www.defense.gov/News/News-Stories/Article/Article/3644027/us-partners-forces-strike-houthi-military-targets-in-yemen/

[66] Ibid.

commercial traffic, military action alone has not eliminated the underlying threat.[67] This makes the United States an important example of the difficulty stronger conventional militaries face when responding to low-cost asymmetric drone warfare by a non-state actor.[68]

Israel

On 19 July 2024, the Houthis launched a long-range UAV from Yemen that struck central Tel Aviv near the U.S. Embassy Branch Office, killing one Israeli civilian and injuring several others. Israeli authorities assessed that the UAV was an Iranian-made Samad-3 that had been modified to extend its range, demonstrating the Houthis' ability to conduct long-distance drone attacks against targets well beyond the Red Sea. [69] The attack was significant because it showed that drone capabilities were not limited to maritime targets and could threaten population centres and critical areas in Israel.[70]Following the attack, Prime Minister Benjamin Netanyahu stated that Israel would respond against those attacking it and authorised strikes on Houthi targets in Yemen, while emphasising that Israel had intercepted hundreds of previous projectiles with its allies.

In response, on 20 July 2024, Israel carried out Operation Outstretched Arm against Houthi-controlled areas around the port of Hudaydah in Yemen. Israeli aircrafts struck fuel-storage facilities, a refinery, a power station and port infrastructure.[71] The strikes reportedly killed several people and wounded dozens, while the damage to Hudaydah's infrastructure raised concerns about its effects on the movement of fuel, food and humanitarian supplies because the port is an important entry point for Yemen.[72] The exchange demonstrated the escalatory nature of drone warfare: a relatively low-cost UAV attack resulted in a long-range Israeli military response against infrastructure in Yemen, extending the conflict geographically and increasing the risk of further regional escalation.[73]

Houthi movement

The Houthi movement, also known as Ansar Allah, is a Yemeni non-state armed group that has developed significant asymmetric capabilities, including drones, ballistic missiles and cruise

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[67] U.S. Department of Defense. (2024). *Senior defense official background briefing on this week's Gulf Cooperation Council meetings*. U.S. Department of Defense. https://www.defense.gov/News/Transcripts/Transcript/Article/3784178/transcript-of-senior-defense-official-background-briefing-on-this-weeks-gulf-co/

[68] Ibid.

[69] Congressional Research Service. (2024). *Israel and Houthis in Yemen: Attacks Highlight Regional Conflict Risks*. Congress.gov. https://crsreports.congress.gov/product/pdf/IN/IN12391

[70] Ibid.

[71] Israel Defense Forces. (2024). *Summary of Operation "Outstretched Arm"*. Israel Defense Forces. https://www.idf.il/en/mini-sites/idf-press-releases-israel-at-war/july-24-pr/summary-of-operation-outstretched-arm/

[72] Congressional Research Service. (2024). *Israel and Houthis in Yemen: Attacks Highlight Regional Conflict Risks*. Congress.gov. https://crsreports.congress.gov/product/pdf/IN/IN12391

[73] Ibid.

missiles.[74] After taking control of Sana'a in 2014, the group continued to expand its military capabilities and later used these systems against targets in Saudi Arabia, Israel and the Red Sea.[75] Since November 2023, the Houthis have repeatedly targeted commercial and military vessels in the Red Sea and Gulf of Aden, using drones and missiles as part of a campaign they have linked to the war in Gaza.[76] These attacks have demonstrated how a non-state actor can use relatively inexpensive unmanned technologies to threaten commercial shipping and impose costs on militarily stronger states.[77] These high economic costs on states and commercial shipping are disproportionate, as defending against or responding to low-cost weapons can require significantly greater financial resources.By August 2024, UN investigators had recorded 134 attacks from Houthi-controlled areas against merchant vessels and U.S. and UK warships, with at least four seafarers killed.[78] The Houthi movement is therefore central to this study because its actions provide a clear example of how drone warfare can enhance the capabilities of a non-state actor and create effects far beyond the immediate battlefield.[79]

Combined Maritime Forces

The Combined Maritime Forces are a multinational naval partnership. The Combined Maritime Forces (CMF) are particularly relevant to the topic because their role is to protect maritime security in the Red Sea and respond to threats from non-state actors. The Combined Maritime Forces (CMF) protect maritime security in the Red Sea by conducting patrols, monitoring shipping routes and responding to threats such as drone and missile attacks against commercial vessels. Combined Task Force 153 (CTF 153), established in 2022, focuses specifically on the Red Sea, Bab al-Mandeb Strait and Gulf of Aden. [80]During the Houthi attacks on commercial shipping, CTF 153 led Operation Prosperity Guardian, coordinating an international maritime presence intended to protect freedom of navigation and commercial shipping. [81] Its forces also helped monitor maritime activity and coordinate responses to threats against merchant vessels. These actions helped improve surveillance

[74] Council on Foreign Relations. (2026). *Iran's Support of the Houthis: What to Know*. Council on Foreign Relations. https://www.cfr.org/articles/irans-support-houthis-what-know

[75] Council on Foreign Relations. (2026). *Conflict in Yemen and the Red Sea*. Council on Foreign Relations. https://www.cfr.org/global-conflict-tracker/conflict/war-yemen

[76] Council on Foreign Relations. (2025). *The Siege of the Red Sea*. Council on Foreign Relations. https://www.cfr.org/articles/siege-red-sea

[77] Ibid.

[78] United Nations Security Council. (2024). *Letter dated 9 October 2024 from the Panel of Experts on Yemen addressed to the President of the Security Council*. United Nations. https://digitallibrary.un.org/record/4079905/files/S_2024_731-EN.pdf

[79] Ibid.

[80] Combined Maritime Forces. (2022). *CTF 153: Red Sea Maritime Security*. Combined Maritime Forces. https://combinedmaritimeforces.com/ctf-153-red-sea-maritime-security/

[81] U.S. Department of Defense. (2024). *U.S., partners' forces strike Houthi military targets in Yemen*. U.S. Department of Defense. https://www.defense.gov/News/News-Stories/Article/Article/3644027/us-partners-forces-strike-houthi-military-targets-in-yemen/

and provide a military presence to protect commercial shipping, although the continued Houthi attacks show that multinational naval operations have reduced but not eliminated the wider threat.[82] In 2025, CTF 153 also tested the use of uncrewed surface vessels for continuous surveillance of the Red Sea, showing how unmanned technology is increasingly being used not only by the Houthis but also by international forces responding to the conflict.[83]

UN Documents and International Agreements

UN Security Council Resolution 2722

UN Security Council Resolution 2722 was adopted on 10 January 2024 in response to the Houthi attacks on merchant and commercial vessels in the Red Sea.[84] The resolution demanded that the Houthis immediately stop their attacks and release the commercial cargo ship Galaxy Leader and its crew.[85] It also reaffirmed that merchant and commercial vessels have navigational rights and freedoms under international law and recognised the right of states to defend their vessels from attacks.[86] The resolution is particularly relevant to drone warfare because the attacks involved the use of drones and missiles by a non-state actor to threaten international shipping. By addressing the attacks as a matter of international peace, maritime security and freedom of navigation, Resolution 2722 demonstrates how the use of unmanned systems in the Red Sea developed from a regional security issue into an international concern.[87]

UN Security Council Resolution 2739 (2024)

UN Security Council Resolution 2739 was adopted on 27 June 2024, as Houthi attacks against merchant and commercial vessels in the Red Sea continued.[88] The resolution reaffirmed the Security

[82] Combined Maritime Forces. (2025). *CTF153 and USS Stout "Ready Together" for Red Sea Maritime Security*. Combined Maritime Forces. https://www.usff.navy.mil/Press-Room/News-Stories/Article/4074955/ctf153-and-uss-stout-ready-together-for-red-sea-maritime-security/

[83] Combined Maritime Forces. (2025). *CTF 153 Uncrewed Vessel Operations Enhance Red Sea Security*. Combined Maritime Forces. https://combinedmaritimeforces.com/2025/04/07/ctf-153-uncrewed-vessel-operations-enhance-red-sea-security/

[84] United Nations Security Council. (2024). *Resolution 2722 (2024): On attacks by Houthis on merchant and commercial vessels*. United Nations. https://digitallibrary.un.org/record/4033392

[85] United Nations Security Council. (2024). *Resolution 2722 (2024): On attacks by Houthis on merchant and commercial vessels*. United Nations. https://digitallibrary.un.org/record/4033392

[86] United Nations Security Council. (2024). *Resolution 2722 (2024): On attacks by Houthis on merchant and commercial vessels*. United Nations. https://digitallibrary.un.org/record/4033392

[87] United Nations Security Council. (2024). *Resolution 2722 (2024): On attacks by Houthis on merchant and commercial vessels*. United Nations. https://digitallibrary.un.org/record/4033392

[88] United Nations Security Council. (2024). *Resolution 2739 (2024): On attacks by Houthis on merchant and commercial vessels*. United Nations. https://digitallibrary.un.org/record/4053356

Council's demand that the Houthis cease their attacks and release the Galaxy Leader and its crew.[89] It also extended the requirement for the UN Secretary-General to provide monthly reports on Houthi attacks, allowing the Security Council to monitor the continuing threat to maritime security.[90] More importantly, the resolution called for caution and restraint to prevent further escalation and emphasised the need to address the wider conflicts contributing to instability in the region.[91] For the topic of drone warfare and asymmetric conflict, Resolution 2739 is significant because it shows that continued attacks by a non-state actor using drones and missiles had become a sustained international security problem rather than an isolated incident.[92]

Previous Attempts to Solve the issue

Operation Prosperity Guardian

Operation Prosperity Guardian (OPG) was announced by the United States in December 2023 in response to Houthi drone and missile attacks on commercial vessels in the Red Sea.[93] The multinational operation brought together countries including the United Kingdom, Bahrain, Canada, France, Italy, the Netherlands, Norway, and Seychelles to strengthen maritime security and protect freedom of navigation.[94] Its forces worked to detect and intercept incoming drones and missiles, helping protect commercial vessels from attacks,[95] through naval patrols, radar and surveillance systems, helicopters and ship-based air-defence systems. However, continued Houthi attacks, despite the operation, demonstrated the difficulty of completely preventing asymmetric attacks using relatively inexpensive drones and missiles.[96] OPG therefore showed both the importance and limitations of multinational naval forces in responding to modern drone warfare. These limitations were partly addressed through increased intelligence-sharing, stronger coordination between allied

[89] United Nations Security Council. (2024). *Resolution 2739 (2024): On attacks by Houthis on merchant and commercial vessels*. United Nations. https://digitallibrary.un.org/record/4053356

[90] United Nations Security Council. (2024). *Resolution 2739 (2024): On attacks by Houthis on merchant and commercial vessels*. United Nations. https://digitallibrary.un.org/record/4053356

[91] Ibid.

[92] Ibid.

[93] U.S. Department of Defense. (2023). *Statement from Secretary of Defense Lloyd J. Austin III on ensuring freedom of navigation in the Red Sea*. U.S. Department of Defense. https://www.defense.gov/News/Releases/Release/Article/3621110/statement-from-secretary-of-defense-lloyd-j-austin-iii-on-ensuring-freedom-of-n/

[94] U.S. Department of Defense. (2023). *Ryder gives more detail on how Operation Prosperity Guardian will work*. U.S. Department of Defense. https://www.defense.gov/News/News-Stories/Article/article/3624836/ryder-gives-more-detail-on-how-operation-prosperity-guardian-will-work/

[95] U.S. Department of Defense. (2024). *NAVCENT commander discusses Operation Prosperity Guardian*. U.S. Department of Defense. https://www.defense.gov/News/Transcripts/Transcript/Article/3631484/navcent-commander-vice-admiral-brad-cooper-holds-an-off-camera-on-the-record-pr/

[96] Ibid.

naval forces, additional air-defence systems, and strikes against Houthi weapons and launch infrastructure.

Regional Programme for Maritime Security in the Red Sea

The Regional Programme for Maritime Security in the Red Sea Area was launched in February 2021 with funding from the European Union to strengthen maritime security capabilities in the region.[97] The programme supports countries including Djibouti, Ethiopia, Somalia, Sudan and Yemen by working with international organisations such as the United Nations, the International Maritime Organization (IMO), and the African Union (AU). The goal is to improve maritime, port and land-based law-enforcement capabilities through training, technical assistance, equipment, information-sharing, and capacity-building programmes for maritime and law-enforcement personnel.[98] Its importance has increased as regional instability and threats to freedom of navigation have affected shipping through the Red Sea.[99] The programme therefore represents a long-term preventative approach, focusing on strengthening regional institutions and maritime-security capacity rather than responding only after attacks occur.[100] The programme has been effective in strengthening regional maritime-security capacity through training, improved information-sharing and greater cooperation between participating states, although its preventative impact is more difficult to measure because it aims to reduce vulnerabilities over the long term rather than directly prevent individual attacks. This is relevant to the drone-warfare issue because stronger surveillance, coordination and maritime-security institutions can improve the ability of regional states to identify and respond to emerging threats to commercial shipping.[101]

Djibouti Code of Conduct / Jeddah Amendment

The Djibouti Code of Conduct, adopted in 2009, brought regional states together, including 21 states from the Western Indian Ocean, Gulf of Aden and Red Sea regions together to strengthen cooperation against piracy and armed robbery at sea. These included Djibouti, Ethiopia, Kenya, Madagascar, Maldives, Seychelles, Somalia, Sudan, Tanzania and Yemen, among others.to combat piracy and armed robbery against ships in the western Indian Ocean and Gulf of Aden.[102] In 2017, the Jeddah Amendment expanded this cooperation to cover broader maritime-security threats and called for

[97] International Maritime Organization. (n.d.). *Red Sea Project*. International Maritime Organization. https://www.imo.org/en/ourwork/security/pages/redseaproject.aspx

[98] International Maritime Organization. (n.d.). *Red Sea Project*. International Maritime Organization. https://www.imo.org/en/ourwork/security/pages/redseaproject.aspx

[99] Ibid.

[100] Ibid.

[101] Ibid.

[102] International Maritime Organization. (n.d.). *Djibouti Code of Conduct*. International Maritime Organization. https://www.imo.org/en/ourwork/security/pages/dcoc.aspx

greater information-sharing, joint action and national and regional capacity-building.[103] In 2017, the Jeddah Amendment expanded this cooperation to cover broader maritime-security threats such as human trafficking, illegal fishing, maritime terrorism, drug trafficking, and other forms of transnational organised crime at sea. The framework is relevant to the Red Sea because it provides a regional mechanism for improving maritime security and responding to emerging threats to shipping.[104] In 2023, participating states also agreed to develop a common regional maritime-security strategy and established working groups addressing new and emerging maritime threats, port and ship security, and information-sharing.[105] Although the framework was not created specifically to address drone attacks, these mechanisms can strengthen regional preparedness and coordination against newer forms of maritime threats. Although the framework has improved regional coordination, information-sharing and maritime-security capacity, its effectiveness against emerging threats such as drone warfare remains limited, as demonstrated by the continued Houthi attacks in the Red Sea.[106]

Possible Solutions

Improving air and drone defence systems

Improving air and drone defence systems is an important solution because asymmetric warfare has demonstrated that relatively inexpensive UAVs can threaten commercial vessels and military forces in the Red Sea.[107] States could invest in advanced ship-based systems capable of detecting, tracking and intercepting drones before they reach their targets, while also improving radar, electronic warfare and other counter-UAV technologies.[108] This could be achieved through investment in naval air-defence systems, upgraded radar and sensor networks, electronic-warfare equipment, and specialised counter-UAV systems, alongside regular training and information-sharing between allied naval forces.NATO has emphasised the importance of developing new technologies, strengthening interoperability between allied forces and working with the private sector to address emerging security threats.[109]

[103] Ibid.

[104] International Maritime Organization. (n.d.). *Red Sea area*. International Maritime Organization. https://www.imo.org/en/mediacentre/hottopics/pages/red-sea.aspx

[105] International Maritime Organization. (2023). *Strategic approach to implementing Djibouti Code of Conduct on maritime security*. International Maritime Organization. https://www.imo.org/en/mediacentre/pages/whatsnew-1986.aspx

[106] International Maritime Organization. (n.d.). *Red Sea area*. International Maritime Organization. https://www.imo.org/en/mediacentre/hottopics/pages/red-sea.aspx

[107] Haugstvedt, H. (2024). *An Update on Violent Non-State Actors' Use of Armed UAVs*. Perspectives on Terrorism. https://www.jstor.org/stable/27301123

[108] NATO. (2024). *NATO's approach to countering unmanned aircraft systems*. NATO. https://www.nato.int/

[109] Ibid.

Applying these principles to the Red Sea could help states share detection systems, coordinate responses and develop more cost-effective methods of countering drones.[110] However, maintaining effective cooperation can be difficult because countries have different military capabilities, technologies and defence systems, making interoperability and information-sharing essential to an effective regional response. This could be addressed through joint military exercises, shared communication standards, compatible defence systems, and real-time intelligence-sharing, allowing states with different capabilities to coordinate their responses more effectively.[111]

Strengthening maritime security

Strengthening maritime security through greater international naval cooperation, surveillance and information-sharing could reduce the impact of drone and missile attacks in the Red Sea. The Combined Maritime Forces (CMF), which brings together naval forces from multiple countries, can increase joint patrols and maritime surveillance around vulnerable areas such as the Bab el-Mandeb Strait, allowing threats to commercial vessels to be detected and responded to more quickly. Safety could be further ensured through continuous surveillance, coordinated naval escorts, rapid communication between vessels and naval forces, and clear procedures for responding to detected drone or missile threats. The International Maritime Organisation (IMO) can complement these efforts by coordinating information-sharing between governments, maritime authorities and shipping companies and by supporting internationally agreed measures for safer navigation.[112] Improved early-warning systems are particularly important because detecting drones and missiles before they reach commercial vessels gives naval forces more time to intercept them and allows ships to take protective measures.[113] Greater cooperation could also reduce duplication between different naval missions and create a more coordinated regional response to attacks.[114] However, maritime security alone cannot eliminate the underlying Houthi threat, meaning that naval protection should be combined with diplomatic efforts to address the wider conflict in Yemen.[115]

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[110] NATO. (2023). *Emerging and disruptive technologies*. NATO. https://www.nato.int/cps/en/natohq/topics_184303.htm

[111] Ibid.

[112] Combined Maritime Forces. (n.d.). *Combined Task Force 153: Red Sea maritime security*. Combined Maritime Forces. https://combinedmaritimeforces.com/ctf-153-red-sea-maritime-security/

[113] International Maritime Organization. (n.d.). *Red Sea*. International Maritime Organization. https://www.imo.org/en/mediacentre/hottopics/pages/red-sea.aspx

[114] Combined Maritime Forces. (n.d.). *Combined Task Force 153: Red Sea maritime security*. Combined Maritime Forces. https://combinedmaritimeforces.com/ctf-153-red-sea-maritime-security/

[115] United Nations. (2023). *Special Envoy for Yemen welcomes parties' commitment to set of measures to implement ceasefire, improve living conditions and engage in political process*. United Nations. https://www.un.org/sg/en/content/sg/press-encounter/2023-12-23/special-envoy-yemen-welcomes-parties%E2%80%99-commitment-set-of-measures-implement-ceasefire-improve-living-conditions-and-engage-political-process

Establishing a Regional Maritime Security Coordination Body

A stronger solution would be to establish a permanent regional maritime-security coordination body for the Red Sea, bringing together Red Sea states, international naval forces, the IMO and commercial shipping representatives. The body could create a shared system for reporting and tracking drone and missile threats, allowing information about attacks, suspicious activity and vulnerable shipping routes to be distributed quickly between participating states.

The IMO already supports regional cooperation and information-sharing on maritime security, while existing initiatives such as the Djibouti Code of Conduct demonstrate that states in the region can cooperate through shared maritime-security mechanisms.[116]A permanent coordination body could build on these existing structures rather than creating an entirely separate system, improving communication between naval forces and commercial shipping companies and allowing earlier warnings to be issued to vessels operating in high-risk areas.[117] This could reduce the vulnerability of commercial vessels to Houthi drone attacks by making threat information available more quickly and coordinating defensive responses across the region.[118]

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[116] International Maritime Organization. (n.d.). *Red Sea*. International Maritime Organization. https://www.imo.org/en/mediacentre/hottopics/pages/red-sea.aspx

[117] International Maritime Organization. (n.d.). *Djibouti Code of Conduct*. International Maritime Organization. https://www.imo.org/en/ourwork/security/pages/dcoc.aspx

[118] International Maritime Organization. (n.d.). *Red Sea*. International Maritime Organization. https://www.imo.org/en/mediacentre/hottopics/pages/red-sea.aspx

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